Ritual Roast Resource Configuration

Below the specs are tabulate for some of the resources that will be created.

VPC Configuration

Component	Details
VPC CIDR	10.16.0.0/16
Internet Gateway	Attached to VPC
NAT Gateway	1 NAT Gateway in Public Subnet 1 (10.16.0.0/20)
Public Subnets	10.16.0.0/20 (eu-west-2a), 10.16.16.0/20(eu-west-2b)
App Subnets (Private)	10.16.64.0/20(eu-west-2a), 10.16.80.0/20 (eu-west-2b)
Data Subnet(Private)	10.16.192.0/20 (eu-west-2a), 10.16.208.0/20 (eu-west-2b)

Subnets configuration

Public subnets

Subnet Name	IPv4 CIDR	AZ
PublicSubnet-01-2a	10.16.0.0/20	eu-west-2a
PublicSubnet-02-2b	10.16.16.0/20	eu-west-2b
PublicSubnet-03-2c	10.16.32.0/20	eu-west-2c

Note the IPv4 CIDR 10.16.48.0/20 is not used here. Lets assume that is reserved for future AZ

Web Subnets

Subnet Name	IPv4 CIDR	AZ
WebSubnet-01-2a	10.16.64.0/20	eu-west-2a
WebSubnet-02-2b	10.16.80.0/20	eu-west-2b
WebSubnet-03-2c	10.16.96.0/20	eu-west-2c

Note the IPv4 CIDR 10.16.112.0/20 is not used here. Lets assume that is reserved for future AZ

App Subnets(Applications future hold)

Subnet Name	IPv4 CIDR	AZ
AppSubnet-01-2a	10.16.128.0/20	eu-west-2a
AppSubnet-02-2b	10.16.144.0/20	eu-west-2b
AppSubnet-03-2c	10.16.160.0/20	eu-west-2c

Note the IPv4 CIDR 10.16.176.0/20 is not used here. Lets assume that is reserved for future AZ

Database Subnets

Subnet Name	IPv4 CIDR	AZ
DataSubnet-01-2a	10.16.192.0/20	eu-west-2a
DataSubnet-02-2b	10.16.208.0/20	eu-west-2b
DataSubnet-03-2c	10.16.224.0/20	eu-west-2c

Note the IPv4 CIDR 10.16.240.0/20 is not used here. Lets assume that is reserved for future AZ

Route tables

Table Name	Route
Public Route Table	0.0.0.0/0 ---> Internet Gateway
Private Route Table (Main)	0.0.0.0/0 ---> NAT Gateway

Main route table

Destination	Target	Status
10.16.0.0/16	Local	Active
0.0.0.0/0	NATGW	Active

Public Route Table

Destination	Target	Status
10.16.0.0/16	Local	Active
0.0.0.0/0	IGW	Active

Security Groups Settings

Security Group	Inbound Rules
LoadBalancer-SG	Allow HTTP (port 80) from 0.0.0.0/0
Web-App-SG	Allow TCP 5000 and HTTP (port 80) from LoadBalancer-SG
Database-SG	Allow TCP 3306 from Web-App-SG and from itself (for Secrets Manager Lambda rotation)

RDS MySQL Configuration

Component	Details
Engine	MySQL
Deployment	Multi-AZ (eu-west-2a and eu-west-2b)
Subnet Group	Includes data subnets (10.16.192.0/20, 10.16.208.0/20)
Storage Type	General Purpose SSD - 20GB Allocated Space
Secrets Manager	Stores RDS Credentials
Rotation	Enabled, every 7 days

S3 Bucket Configuration

Component	Details
Purpose	Hosting Flask application source code
Bucket Policy	Only accessible to EC2 via IAM Role
StorageClass	Standard
Versioning	Enabled
Bucket Replication	N/A

EC2 Instance Configuration

Component	Details
AMI	Amazon Linux 2023
Instance Profile	IAM Role with S3, SSM, Secrets Manager permissions
Permissions	AmazonS3FullAccess, AmazonSSMManagedInstanceCore, SecretsManagerReadWrite

IAM Role for EC2

Permissions	Purpose
AmazonS3FullAccess	Fetch application code (Restrict to bucket)
AmazonSSManagedInstanceCore	SSM Session Manager access
SecretManagerReadWrite(GetSecretValue)	Fetch DB credentials securely

Application LoadBalancer Configuration

Component	Details
Type	Internet-facing
Subnets	Public Subnets 1 & 2
Security Group	LoadBalancer-SG
Target Group	Targets on port 5000 (EC2s) - To be configured by ASG
Listener	HTTP listener on port 80 forwards to target group
Health Checks	TCP/5000 or HTTP endpoint (e.g., /)

AutoScaling Group Configuration

Component	Details
Launch Template	Includes AMI, IAM Role, and UserData
UserData Script	- Install Python, Flask, etc - Download source from S3 - Run Flask app
Scaling Policy	Target Tracking
Instance Count	Min:2, Desired: 2, Max: 4
Subnets	App/Web Subnets(Private): 10.16.64.0/20 10.16.80.0/20